Pratham Kumar

Artificial Intelligence / Machine Learning Engineer

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WORK EXPERIENCE

AI and Ml Engineer

Visioapps Technology Jan 2025 - Present

- **Custom Model Development (LLMs, NER, Intent Classification)**
- Designed and implemented Intent Recognition and Named Entity Recognition (NER) models using SpaCy and BERT, achieving 98% accuracy for financial domain text understanding.
- Fine tuned and optimized Large Language Models (LLMs), combined prompt engineering, model tuning, and multi step workflows to improve contextual consistency and reduce hallucinations across financial use cases.
- Improved Louie Voice Software accuracy from 90% to 95%+, enhancing reliable voice driven task execution.
- Reduced manual effort for internal teams by 30% through automation driven model improvements and optimized NLP pipelines.
- **RAG & Information Retrieval Systems**
- Built a domain-specific Question Answering system using Retrieval-Augmented Generation (RAG) for banking customer data.
- Developed a document ingestion pipeline for PDFs, forms, and unstructured text using LangChain loaders with optimized chunking strategies.
- Implemented FAISS based semantic search for high-speed vector retrieval across banking documents.
- Combined retrieved context with LLMs to generate accurate, context aware responses for client queries.
- **ML Pipeline & Production Optimization**
- Gained working knowledge of FastAPI/Flask based ML service deployment and AWS workflows, collaborating with senior engineers on integration and testing activities.
- Applied optimization techniques to reduce inference time, improving overall model latency by 20-30% and achieving up to 50% faster processing in Speech to Text (STT) components used within the software.

PROJECTS

House Prices- Advances Regression Techniques may 2024 - jun 2024

- Ranked in the top 2% on leaderboards.
- Developed predictive models using Random Forest, Gradient Boosting (XGBoost and LightGBM), and pandas for data manipulation.
- Conducted extensive data preprocessing, feature engineering, and outlier detection to improve model accuracy.
- Utilized advanced techniques such as hyperparameter tuning and cross validation to optimize model performance.
- Achieved a competitive Root Mean Squared Error (RMSE) score of **0.11790** on the test dataset.

SKILLS

Frameworks: Python, NumPy, Pandas, PyTorch, TensorFlow, Keras, scikit-learn, Neural Networks (CNN/RNN/Transformers)
Domain: Fintech, Generative AI (GenAI), MLOps, Automation workflows, Predictive Analytics, Research
Data: SQL, Data Cleaning, Preprocessing, EDA, Feature Engineering, Model Evaluation, Visualization
Tools & Platforms: Jupyter Notebook, Git, Docker, N8N, AWS, Jira, PySpark

EDUCATION

B.Tech in CSE(AI and ML)

Guru Jambheshwar University, Hisar 2021 - 2025

Certifications

- Data Analysis Virtual Internship - CSRBOX + IBM SkillsBuild
- Artificial Intelligence Virtual Internship - CSRBOX + IBM SkillsBuild
- Python Masterclass Course - Udemy
- LLM Engineering Course - Udemy